TEXAS INSTRUMENTS

SN5400, SN54LS00, SN54S00
SN7400, SN74LS00, SN74S00

SDLS025D – DECEMBER 1983 – REVISED MAY 2017

SNx400, SNx4LS00, and SNx4S00 Quadruple 2-Input Positive-NAND Gates

1 Features

- Package Options Include:
  - Plastic Small-Outline (D, NS, PS)
  - Shrink Small-Outline (DB)
  - Ceramic Flat (W)
  - Ceramic Chip Carriers (FK)
  - Standard Plastic (N)
  - Ceramic (J)
- Also Available as Dual 2-Input Positive-NAND Gate in Small-Outline (PS) Package
- Inputs Are TTL Compliant; V_{IH} = 2 V and V_{IL} = 0.8 V
- Inputs Can Accept 3.3-V or 2.5-V Logic Inputs
- SN5400, SN54LS00, and SN54S00 are Characterized For Operation Over the Full Military Temperature Range of –55ºC to 125ºC

2 Applications

- AV Receivers
- Portable Audio Docks
- Blu-Ray Players
- Home Theater
- MP3 Players or Recorders
- Personal Digital Assistants (PDAs)

3 Description

The SNx4xx00 devices contain four independent, 2-input NAND gates. The devices perform the Boolean function $Y = \overline{A \cdot B}$ or $Y = \overline{A} + \overline{B}$ in positive logic.

Device Information(1)

PART NUMBER	PACKAGE	BODY SIZE (NOM)
SN74LS00DB	SSOP (14)	6.20 mm × 5.30 mm
SN7400D, SN74LS00D, SN74S00D	SOIC (14)	8.65 mm × 3.91 mm
SN74LS00NSR	PDIP (14)	19.30 × 6.35 mm
SNJ5400J, SNJ54LS00J, SNJ54S00J	CDIP (14)	19.56 mm × 6.67 mm
SNJ5400W, SNJ54LS00W, SNJ54S00W	CFP (14)	9.21 mm × 5.97 mm
SN54LS00FK, SN54S00FK	LCCC (20)	8.89 mm × 8.89 mm
SN7400NS, SN74LS00NS, SN74S00NS	SO (14)	10.30 mm × 5.30 mm
SN7400PS, SN74LS00PS	SO (8)	6.20 mm × 5.30 mm

(1) For all available packages, see the orderable addendum at the end of the data sheet.

Logic Diagram, Each Gate (Positive Logic)



An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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4 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision C (November 2016) to Revision D **Page**

- Changed *Typical Application* Diagram see *Application and Implementation* section ... 1

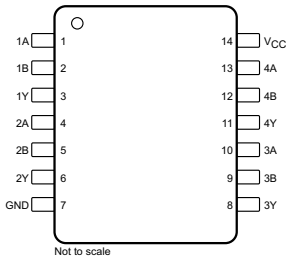
Changes from Revision B (October 2003) to Revision C **Page**

- Added *ESD Ratings* table, *Feature Description* section, *Device Functional Modes*, *Application and Implementation Power Supply Recommendations* section, *Layout* section, *Device and Documentation Support* section, and *Mechanical, Packaging, and Orderable Information* section. ... 1
- Changed *Ordering Information* table to *Device Comparison Table*; see *Package Option Addendum* at the end of the data sheet ... 1
- Changed Package thermal impedance, $R_{\theta JA}$, values in *Thermal Information* table From: 86°C/W To: 90.9°C/W (D), From: 96°C/W To: 102.8°C/W (DB), From: 80°C/W To: 54.8°C/W (N), and From: 76°C/W To: 89.7°C/W (NS) ... 6

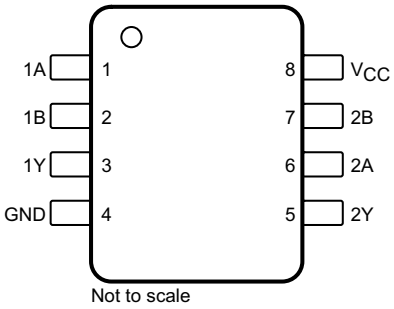
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5 Pin Configuration and Functions

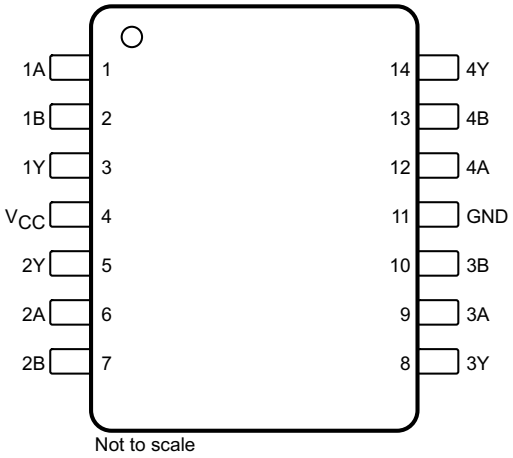
SN5400 J, SN54xx00 J and W, SN74x00 D, N, and NS, or SN74LS00 D, DB, N, and NS Packages
14-Pin CDIP, CFP, SOIC, PDIP, SO, or SSOP
Top View



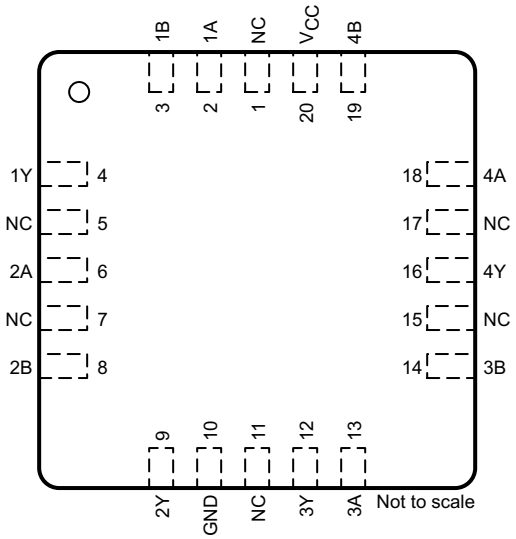
SN5400 W Package
14-Pin CFP
Top View



SN74xx00 PS Package
18-Pin SO
Top View



SN54xx00 FK Package
20-Pin LCCC
Top View



Pin Functions

PIN					I/O	DESCRIPTION
NAME	CDIP, CFP, SOIC, PDIP, SO, SSOP	SO (SN74xx00)	CFP (SN5400)	LCCC		
1A	1	1	1	2	I	Gate 1 input
1B	2	2	2	3	I	Gate 1 input
1Y	3	3	3	4	O	Gate 1 output
2A	4	6	6	6	I	Gate 2 input
2B	5	7	7	8	I	Gate 2 input
2Y	6	5	5	9	O	Gate 2 output
3A	10	—	9	13	I	Gate 3 input
3B	9	—	10	14	I	Gate 3 input

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Pin Functions (continued)

PIN					I/O	DESCRIPTION
NAME	CDIP, CFP, SOIC, PDIP, SO, SSOP	SO (SN74xx00)	CFP (SN5400)	LCCC		
3Y	8	—	8	12	O	Gate 3 output
4A	13	—	12	18	I	Gate 4 input
4B	12	—	13	19	I	Gate 4 input
4Y	11	—	14	16	O	Gate 4 output
GND	7	4	11	10	—	Ground
NC	—	—	—	1, 5, 7, 11, 15, 17	—	No connect
V_{CC}	14	8	4	20	—	Power supply

6 Specifications

6.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

		MIN	MAX	UNIT
Supply voltage, V_{CC}(2)			7	V
Input voltage	SNx400 and SNxS400		5.5	V
	SNx4LS00		7	
Junction temperature, T_J			150	°C
Storage temperature, T_{stg}		–65	150	°C

(1) Stresses beyond those listed under *Absolute Maximum Ratings* may cause permanent damage to the device. These are stress ratings only, which do not imply functional operation of the device at these or any other conditions beyond those indicated under *Recommended Operating Conditions*. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
(2) Voltage values are with respect to network ground terminal.

6.2 ESD Ratings: SN74LS00

			VALUE	UNIT
$V_{(ESD)}$	Electrostatic discharge	Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)	±500	V
		Charged-device model (CDM), per JEDEC specification JESD22-C101(2)	±2000	

(1) JEDEC document JEP155 states that 500-V HBM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 500-V HBM is possible with the necessary precautions.
(2) JEDEC document JEP157 states that 250-V CDM allows safe manufacturing with a standard ESD control process. Manufacturing with less than 250-V CDM is possible with the necessary precautions. Pins listed as ±2000 V may actually have higher performance. ESD Tested on SN74LS00N package.

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
V_{CC}	Supply voltage	SN54xx00	4.5	5	5.5	V
		SN74xx00	4.75	5	5.25	
V_{IH}	High-level input voltage		2			V
V_{IL}	Low-level input voltage	SNx400, SN7LS400, and SNx4S00			0.8	V
		SN54LS00			0.7	
I_{OH}	High-level output current	SN5400, SN54LS00, and SN74LS00			–0.4	mA
		SNx4S00			–1	
I_{OL}	Low-level output current	SNx400			16	mA
		SN5LS400			4	
		SN7LS400			8	
		SNx4S00			20	

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Recommended Operating Conditions (continued)

over operating free-air temperature range (unless otherwise noted)

			MIN	NOM	MAX	UNIT
T_A	Operating free-air temperature	SN54xx00	–55		125	°C
		SN74xx00	0		70	

6.4 Thermal Information

THERMAL METRIC[1][2]		SN74LS00				UNIT
		D (SOIC)	DB (SSOP)	N (PDIP)	NS (SO)	
		14 PINS	14 PINS	14 PINS	14 PINS	
$R_{\theta JA}$	Junction-to-ambient thermal resistance	90.9	102.8	54.8	89.7	°C/W
$R_{\theta JC(top)}$	Junction-to-case (top) thermal resistance	51.9	53.3	42.1	48.1	°C/W
$R_{\theta JB}$	Junction-to-board thermal resistance	48	53.4	34.8	50.1	°C/W
ψ_{JT}	Junction-to-top characterization parameter	18.6	16.5	26.9	16.7	°C/W
ψ_{JB}	Junction-to-board characterization parameter	47.8	52.9	34.7	49.8	°C/W

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application report.
(2) The package thermal impedance is calculated in accordance with JESD 51-7.

6.5 Electrical Characteristics: SNx400

over operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{IK}	V_{CC} = MIN and I_I = –12 mA				–1.5	V
V_{OH}	V_{CC} = MIN, V_{IL} = 0.8 V, and I_{OH} = –0.4 mA		2.4	3.4		V
V_{OL}	V_{CC} = MIN, V_{IH} = 2 V, and I_{OL} = 16 mA			0.2	0.4	V
I_I	V_{CC} = MAX and V_I = 5.5 V				1	mA
I_{IH}	V_{CC} = MAX and V_I = 2.4 V				40	µA
I_{IL}	V_{CC} = MAX and V_I = 0.4 V				–1.6	mA
I_{OS}	V_{CC} = MAX	SN5400	–20		–55	mA
		SN7400	–18		–55	
I_{CCH}	V_{CC} = MAX and V_I = 0 V			4	8	mA
I_{CCL}	V_{CC} = MAX and V_I = 4.5 V			12	22	mA

6.6 Electrical Characteristics: SNx4LS00

over operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS		MIN	TYP	MAX	UNIT
V_{IK}	V_{CC} = MIN and I_I = –18 mA				–1.5	V
V_{OH}	V_{CC} = MIN, V_{IL} = MAX, and I_{OH} = –0.4 mA		2.5	3.4		V
V_{OL}	V_{CC} = MIN and V_{IH} = 2 V	I_{OL} = 4 mA		0.25	0.4	V
		I_{OL} = 8 mA (SN74LS00)		0.35	0.5	
I_I	V_{CC} = MAX and V_I = 7 V				0.1	mA
I_{IH}	V_{CC} = MAX and V_I = 2.7 V				20	µA
I_{IL}	V_{CC} = MAX and V_I = 0.4 V				–0.4	mA
I_{OS}	V_{CC} = MAX		–20		–100	mA
I_{CCH}	V_{CC} = MAX and V_I = 0 V			0.8	1.6	mA
I_{CCL}	V_{CC} = MAX and V_I = 4.5 V			2.4	4.4	mA

6.7 Electrical Characteristics: SNx4S00

over operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
V_{IK}	V_{CC} = MIN and I_I = –18 mA			–1.2	V
V_{OH}	V_{CC} = MIN, V_{IL} = 0.8 V, and I_{OH} = –1 mA	2.5	3.4		V
V_{OL}	V_{CC} = MIN, V_{IH} = 2 V, and I_{OL} = 20 mA			0.5	V
I_I	V_{CC} = MAX and V_I = 5.5 V			1	mA
I_{IH}	V_{CC} = MAX and V_I = 2.7 V			50	µA
I_{IL}	V_{CC} = MAX and V_I = 0.5 V			–2	mA

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Electrical Characteristics: SNx4S00 (continued)

over operating free-air temperature range (unless otherwise noted)

PARAMETER	TEST CONDITIONS	MIN	TYP	MAX	UNIT
I_{OS}	V_{CC} = MAX	–40		–100	mA
I_{CCH}	V_{CC} = MAX and V_I = 0 V		10	16	mA
I_{CCL}	V_{CC} = MAX and V_I = 4.5 V		20	36	mA

6.8 Switching Characteristics: SNx400

V_{CC} = 5 V, T_A = 25°C, and over operating free-air temperature range (unless otherwise noted). See Figure 2.

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH}	A or B	Y	R_L = 400 Ω and C_L = 15 pF		11	22	ns
t_{PHL}					7	15	

6.9 Switching Characteristics: SNx4LS00

V_{CC} = 5 V, T_A = 25°C, and over operating free-air temperature range (unless otherwise noted). See Figure 2.

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH}	A or B	Y	R_L = 2 kΩ and C_L = 15 pF		9	15	ns
t_{PHL}					10	15	

6.10 Switching Characteristics: SNx4S00

V_{CC} = 5 V, T_A = 25°C, and over operating free-air temperature range (unless otherwise noted). See Figure 2.

PARAMETER	FROM (INPUT)	TO (OUTPUT)	TEST CONDITIONS	MIN	TYP	MAX	UNIT
t_{PLH}	A or B	Y	R_L = 280 Ω and C_L = 15 pF		3	4.5	ns
			R_L = 280 Ω and C_L = 50 pF		4.5		
t_{PHL}	A or B	Y	R_L = 280 Ω and C_L = 15 pF		3	5	
			R_L = 280 Ω and C_L = 50 pF		5		

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6.11 Typical Characteristics

C_L = 15 pF

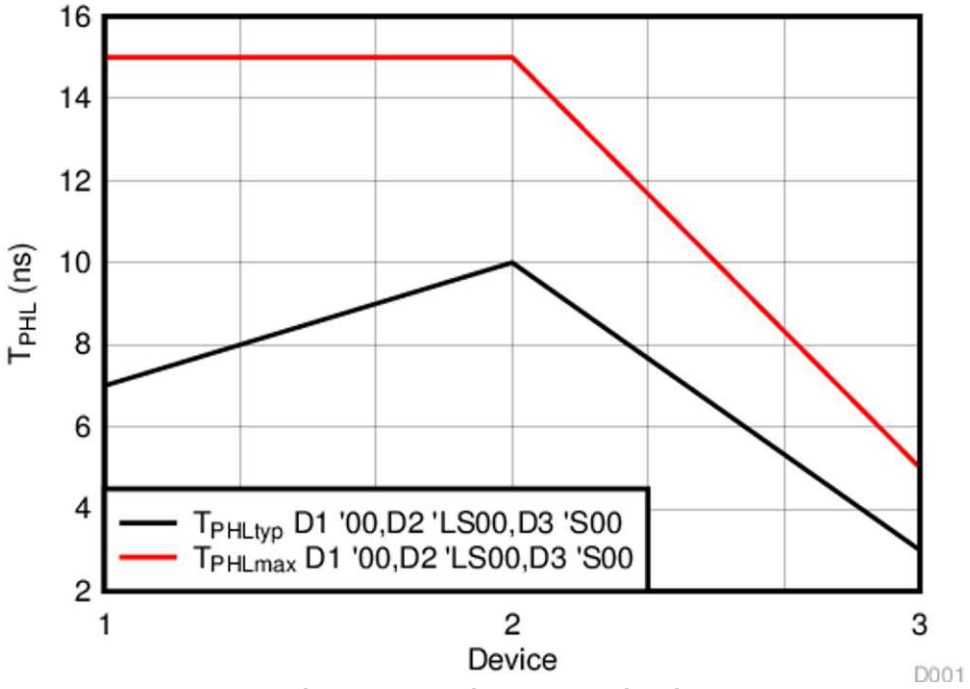


Figure 1. T_{PHL} (Across Devices)

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